Total No. of Questions : 4]

SEAT No. :

PE-557

[Total No. of Pages : 2

[6578]-30

S.E. (Information Technology) (Insem.)

OBJECT ORIENTED PROGRAMMING

(2019 Pattern) (Semester - III) (214444)

Time : 1 Hour] ***[Max. Marks : 30***

Instructions to the candidates:

1) ***Attempt Q1 or Q2, Q3 or Q4***
2) ***Figures to the right indicate full marks.***
3) ***Assume suitable data if necessary.***

Q1) a) Define the need for object-oriented programming. Explain Abstraction and Encapsulation features of OOP. **[5]**

b) Explain the concept of message passing with examples. **[5]**

c) Model a real-world scenario for the 'Fruit' class using an object-oriented paradigm. **[5]**

OR

Q2) a) Write a short note on. **[5]**

- Static and Dynamic Binding
- Information hiding

b) Differentiate between class and object with respect to the following points memory allocation, purpose, content, syntax, and examples. **[5]**

c) What is Polymorphism? Explain the use of Polymorphism. **[5]**

P.T.O.

Q3) a) What are visibility/access modifiers in Java? State the different visibility/access modifiers along with their scope. **[5]**

b) Write and explain the syntax for the method in Java. State the advantages of using methods. **[5]**

c) Write a simple Java program demonstrating the usage of an array of objects. **[5]**

OR

Q4) a) What is Abstract Data Type (ADT)? List the advantages of ADTs. **[5]**

b) Implement a class box to calculate the volume of the box with the following properties

Attributes: width, height, depth

Methods: getDimensions, calculateVolume

Make use of appropriate access specifiers while implementing the class. **[5]**

c) Implement class shape to demonstrate the concept of method overloading. Class defines the methods to calculate the area of different shapes. **[5]**